

CH:4 HIDDEN MARKOV MODELS (HMM) FOR SEQUENTIAL PATTERN CLASSIFICATION

Contents

Markov Model	1
Hidden Markov Model (HMM)	1
Assumptions in HMM	2
Three Main Problems Solved by HMM	3
Continuous Density HMM	6

Markov Model

A Markov Model (or Markov Chain) is a stochastic model that represents a system where the next state depends only on the current state, not on the sequence of events that preceded it.

This is based on the Markov Property:

$$P(X_{t+1} = x | X_t = x_t, X_{t-1} = x_{t-1}, \dots, X_0 = x_0) = P(X_{t+1} = x | X_t = x_t)$$

- **State:** A possible condition or situation in the model
- **Transition:** Movement from one state to another
- **Transition Probability:** Probability of moving from one state to another
- **Initial State Distribution:** Probability distribution over starting states
- **First-Order Markov Assumption:** The probability of the next state depends only on the current state, not the full history.

Components of a Markov Model

1. Set of states: $S = \{s_1, s_2, \dots, s_N\}$
2. Transition probability matrix (A): $A = [a_{ij}],$
 $a_{ij} = P(s_j | s_i)$
3. Initial state distribution (π): $\pi_i = P(\text{initial state is } s_i)$

Markov Chains: are mathematical models that represent systems which move from one state to another, where the probability of each future state depends **only on the current state** and not on the sequence of past states.

Hidden Markov Model (HMM)

A Hidden Markov Model is a statistical model used to represent systems that follow a Markov process with hidden (unobservable) states.

It models a sequence of observations generated by underlying hidden states, where:

- States are not directly visible.
- Each state generates an observable output based on a probability distribution.

Components of an HMM

A Hidden Markov Model is defined by the 5-tuple: $\lambda = (A, B, \pi, N, M)$

- N – Number of hidden states in the model.
- M – Number of distinct observable symbols per state.
- A – Transition probability matrix between hidden states.
- B – Emission probability matrix for observations from states.
- π – Initial probability distribution over the hidden states.

1. States (N):

- A finite set of hidden states: $S = \{S_1, S_2, \dots, S_N\}$
- We don't observe these states directly; they are *hidden*.

2. Initial State Distribution (π): $\pi_i = P(q_1 = S_i)$: Probability the model starts in state S_i .

3. Transition Probability Matrix (A):

- $A = \{a_{ij}\}$, where
 $a_{ij} = P(q_{t+1} = S_j \mid q_t = S_i)$: Probability of going from state i to state j .

4. Observation Symbols (V):

- Set of possible outputs: $V = \{v_1, v_2, \dots, v_M\}$
- These are what we actually observe.

5. Observation Probability Distribution (B):

- $b_i(k) = P(\text{observation} = v_k \mid \text{state} = S_i)$
- Each state has a probability distribution over the observation symbols.

Assumptions in HMM

1. Markov Assumption

Future state depends only on the current state: $P(q_t | q_{t-1}, \dots, q_1) = P(q_t | q_{t-1})$

2. Output Independence Assumption

Observation depends only on current state: $P(O_t | q_t, q_{t-1}, \dots, O_1) = P(O_t | q_t)$

3. For any state in a Markov model, the sum of the probabilities of all outgoing transitions from that state is equal to 1.

4. Stationary Distribution

A distribution π is stationary if:

$$\pi P = \pi \quad \text{and} \quad \sum_i \pi_i = 1$$

This means if the chain starts in π , it remains in π .

5. Stochastic Matrix Conditions:

- All entries are non-negative: $P_{ij} \geq 0$ for all i, j
- Each row sums to 1: $\sum_j P_{ij} = 1$ for each state i

6. State Distribution Update:

The state distribution at the next time step is obtained by multiplying the current state distribution vector π by the state transition matrix A , i.e., $\pi^{(t+1)} = \pi^{(t)} \cdot A$ or $\pi = \pi \cdot A$

This shows how the probability of being in each state evolves over time in a Markov chain.

Three Main Problems Solved by HMM

Problem	Goal	Solution (Algorithm)
Evaluation	Compute the probability of an observation sequence given the model	Forward Algorithm – efficiently calculates the total probability
Decoding	Find the most likely sequence of hidden states for a given observation sequence	Viterbi Algorithm – finds the optimal state path
Learning	Optimize the model parameters (A, B, π) to best fit a given set of observations	Baum-Welch Algorithm – a type of Expectation-Maximization (EM)

Forward Algorithm (Evaluation)

- **Purpose:** To compute the probability of an observation sequence given an HMM.
- **Idea:** Uses dynamic programming to efficiently sum over all possible hidden state paths without listing them explicitly.
- **Key Point:** Avoids exponential complexity by recursively calculating intermediate probabilities.

Viterbi Algorithm (Decoding)

- **Purpose:** To find the most likely sequence of hidden states for a given observation sequence.
- **Idea:** Uses dynamic programming to track the best path (maximum probability) leading to each state at each time step.
- **Key Point:** Selects the path with the **highest probability**, not just total likelihood.

Baum-Welch Algorithm (Learning)

- A special case of the **Expectation-Maximization (EM)** algorithm.
- **Purpose:** Iteratively updates A, B, π to **maximize the likelihood** of the observed sequence.
- **Key Point:** Starts with initial guesses and improves them to maximize the likelihood of the observation data.

Applications of HMMs

- Speech recognition (e.g., Siri, Alexa)
- Part-of-speech tagging (NLP)
- Bioinformatics (gene prediction)
- Gesture recognition
- Financial modelling

Advantages:

- Handles **sequential data** well.
- Based on **solid probability theory**.
- Can model both visible and hidden dynamics.

Disadvantages:

- Requires **large training data**.
- **Assumes independence** of observations (which may not be realistic).
- Can be **computationally expensive** for large state/observation spaces.

Example

Q1. Weather Prediction using Markov Chain (15 marks)

Given a Markov chain with 3 weather states: Sunny, Cloudy, Rainy.

Transition matrix:

$$A = \begin{bmatrix} 0.6 & 0.3 & 0.1 \\ 0.2 & 0.5 & 0.3 \\ 0.1 & 0.4 & 0.5 \end{bmatrix}$$

a) Draw the transition diagram.

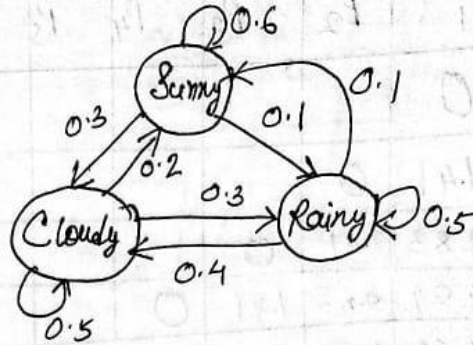
b) Verify that the matrix is stochastic.

c) Find the probability that it rains on Day 3, given it's sunny on Day 1.

d) Find the steady-state probabilities.

Answer

① $A = \begin{bmatrix} 0.6 & 0.3 & 0.1 \\ 0.2 & 0.5 & 0.3 \\ 0.1 & 0.4 & 0.5 \end{bmatrix}$



② for matrix to be stochastic, all elements must be +ve which is true &.

$$\sum P_{ij} = 1 \begin{cases} 0.6 + 0.3 + 0.1 = 1 \\ 0.2 + 0.5 + 0.3 = 1 \\ 0.1 + 0.4 + 0.5 = 1 \end{cases} \text{ true } \therefore \text{Stochastic}$$

③ Prob that it rain on 3rd day given, its sunny in 1st day.

Day 1 (Initial state) = sunny $\rightarrow \pi^{(0)} = [1, 0, 0]$

We go from Day 1 $\rightarrow$ Day 2, then Day 2 $\rightarrow$ Day 3.

$\therefore$ 2 steps, $\therefore \pi^{(2)} = \pi^{(0)} A^2$

$$A^2 = \begin{pmatrix} 0.6 & 0.3 & 0.1 \\ 0.2 & 0.5 & 0.3 \\ 0.1 & 0.4 & 0.5 \end{pmatrix} \begin{pmatrix} 0.6 & 0.3 & 0.1 \\ 0.2 & 0.5 & 0.3 \\ 0.1 & 0.4 & 0.5 \end{pmatrix} = \begin{pmatrix} 0.43 & 0.37 & 0.2 \\ 0.23 & 0.43 & 0.32 \\ 0.19 & 0.43 & 0.38 \end{pmatrix}$$

$$\therefore \pi^{(2)} = \pi^{(0)} A^2 = [1, 0, 0] \begin{pmatrix} 0.43 & 0.37 & 0.2 \\ 0.23 & 0.43 & 0.32 \\ 0.19 & 0.43 & 0.38 \end{pmatrix} = [0.43 \quad 0.37 \quad 0.2]$$

$\therefore$ The probability to rain = 0.2 on 3rd day

③ Steady state vector (π) = $[\pi_1, \pi_2, \pi_3]$

$$\pi \cdot A = \pi$$

$$\Rightarrow [\pi_1, \pi_2, \pi_3] = [\pi_1, \pi_2, \pi_3] \begin{bmatrix} 0.6 & 0.3 & 0.1 \\ 0.2 & 0.5 & 0.3 \\ 0.1 & 0.4 & 0.5 \end{bmatrix}$$

$$\therefore 0.6\pi_1 + 0.2\pi_2 + 0.1\pi_3 = \pi_1$$

$$0.3\pi_1 + 0.5\pi_2 + 0.4\pi_3 = \pi_2$$

$$0.1\pi_1 + 0.3\pi_2 + 0.5\pi_3 = \pi_3$$

$$\therefore \begin{cases} -0.4\pi_1 + 0.2\pi_2 + 0.1\pi_3 = 0 \\ 0.3\pi_1 - 0.5\pi_2 + 0.4\pi_3 = 0 \\ 0.1\pi_1 + 0.3\pi_2 - 0.5\pi_3 = 0 \end{cases} \quad \begin{cases} \text{Also} \\ \pi_1 + \pi_2 + \pi_3 = 1 \end{cases}$$

$$\pi_1 = \frac{0.2\pi_2 + 0.1\pi_3}{0.4} \quad \pi_1 = \frac{0.3\pi_2 - 0.4\pi_3}{0.3}$$

$$\underline{\underline{= 0.5\pi_2 + 0.25\pi_3}} \quad \underline{\underline{0.3\pi_2 + 0.25\pi_3 = \frac{0.3\pi_2 - 0.4\pi_3}{0.3}}}$$

$$\Rightarrow 0.15\pi_3 + 0.075\pi_3 = 0.3\pi_2 - 0.4\pi_3$$

$$\Rightarrow 0.475\pi_3 = 0.35\pi_2 \Rightarrow \boxed{\pi_2 = 1.3571\pi_3}$$

$$\pi_1 + \pi_2 + \pi_3 = 1$$

$$\Rightarrow 0.5(1.3571\pi_3) + 0.25\pi_3 + 1.3571\pi_3 + \pi_3 = 1$$

$$\Rightarrow 0.80355\pi_3 + 1.3571\pi_3 + \pi_3 = 1$$

$$\Rightarrow \boxed{\pi_3 = 0.31639}$$

$$\boxed{\pi_2 = 0.42937}$$

$$\boxed{\pi_1 = 0.25423}$$

$$\therefore \pi = [0.254, 0.429, 0.316]$$

Continuous Density HMM

- In classical (discrete) HMMs, observations come from a **discrete set** of symbols.
- In many real-world problems (like speech recognition, bioinformatics), observations are **continuous-valued vectors** (e.g., acoustic feature vectors).
- **Continuous Density HMM (CD-HMM)** extends HMMs to model continuous observation spaces by using **continuous probability density functions** instead of discrete probability mass functions.

Model Components of a Continuous Density HMM

1. **States (N):** $S = \{S_1, S_2, \dots, S_N\}$ — hidden states
2. **Initial State Distribution (π):** $\pi_i = P(q_1 = S_i)$
3. **Transition Probability Matrix (A):** $A = \{a_{ij}\}$,

$$a_{ij} = P(q_{t+1} = S_j \mid q_t = S_i)$$
4. **Continuous Observations:**
 - Observation at time t is a **real-valued feature vector**: $\mathbf{o}_t = [o_1, o_2, \dots, o_d]$
 - d is the dimensionality of the vector (e.g., 13 for MFCCs in speech)
5. **Observation Probability Density Function (PDF):**
 - Modeled using **Gaussian Mixture Models (GMM)**
 - For state i , the observation likelihood is:

$$b_i(\mathbf{o}) = \sum_{m=1}^M c_{im} \cdot \mathcal{N}(\mathbf{o}; \mu_{im}, \Sigma_{im})$$

- M : Number of Gaussians in the mixture
- c_{im} : Mixture weight for component m
- μ_{im} : Mean vector of component m
- Σ_{im} : Covariance matrix of component m
- $\mathcal{N}$: Multivariate Gaussian function

Multivariate Gaussian PDF Formula

$$\mathcal{N}(\mathbf{o}; \mu, \Sigma) = \frac{1}{(2\pi)^{d/2} |\Sigma|^{1/2}} \exp \left(-\frac{1}{2} (\mathbf{o} - \mu)^T \Sigma^{-1} (\mathbf{o} - \mu) \right)$$

- d = dimension of observation vector $\mathbf{o}$.
- $|\Sigma|$ = determinant of covariance matrix Σ .
- μ = mean vector.
- Σ = covariance matrix (usually diagonal or full).

Question

A CD-HMM has 2 hidden states S_1 and S_2 , each using a **single 2D Gaussian** for observation modeling.

- Observation vector: $\mathbf{o} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$
- Initial distribution: $\pi = [0.6, 0.4]$
- Transition matrix: $A = \begin{bmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{bmatrix}$
- State PDFs: $\mu_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \Sigma_1 = I;$
 $\mu_2 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \Sigma_2 = I$

1. Compute the probability density $b_1(\mathbf{o})$ and $b_2(\mathbf{o})$ using the multivariate Gaussian formula.
2. Using the forward algorithm (first step only), compute the probability of observing vector $\mathbf{o}$ at time $t = 1$.
3. Which state is more likely to have generated the observation at $t = 1$? Justify using calculated values.

$$a) b_i(0) = \sum c_{im} \cdot \mathcal{N}$$

$$\mathcal{N} = \frac{1}{\sqrt{(2\pi)^d |\Sigma|}} e^{-\frac{1}{2} (0-\mu)^T \Sigma^{-1} (0-\mu)}$$

$$\therefore \text{given } \Sigma = I$$

$$\therefore b_i(0) = \frac{1}{\sqrt{(2\pi)^d}} e^{-\frac{1}{2} (0-\mu)^T (0-\mu)} \quad d=2$$

$$= \frac{1}{2\pi} e^{-\frac{1}{2} (0-\mu)^T (0-\mu)}$$

$$\therefore b_1(0) = \frac{1}{2\pi} e^{-\frac{1}{2} \begin{bmatrix} 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \end{bmatrix}} \quad \left| \begin{bmatrix} 2 \\ 1 \end{bmatrix} - \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \right.$$

$$= \frac{1}{2\pi} = 0.1592$$

$$b_2(0) = \frac{1}{2\pi} e^{-\frac{1}{2} \begin{bmatrix} 2 \\ 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix}} \quad \left| \begin{bmatrix} 2 \\ 1 \end{bmatrix} - \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \right.$$

$$= \frac{1}{2\pi} e^{-\frac{5}{2}} = 0.0131$$

$$b) q_i(1) = \pi_i \times b_i(0)$$

$$q_1(1) = 0.6 \times 0.1592 = 0.09552$$

$$q_2(1) = 0.4 \times 0.0131 = 0.00524$$

$$c) \text{ as } \alpha_1(1) > \alpha(2)$$

S_1 is more likely to generate 0 at $t=1$

Feature	Discrete HMM	Continuous HMM
Definition	HMM where observations are from a finite set of symbols	HMM where observations are continuous-valued vectors
Observation type	Discrete symbols (e.g., letters, tokens)	Real-valued feature vectors (e.g., audio/sensor features)
Output distribution	Probability Mass Function (PMF)	Probability Density Function (PDF), often Gaussian or Gaussian Mixture Model (GMM)
Observation space	Finite set (e.g., {a, b, c})	Continuous domain $\mathbb{R}^n$
Emission model	Observation probability table (lookup-based)	Uses statistical parameters (mean, variance, etc.)
Complexity	Simpler to implement and compute	More complex, requires numerical computation
Application examples	Text classification, POS tagging, DNA sequencing	Speech recognition, gesture detection, EEG signal modeling
Parameter estimation	Based on symbol counts/frequencies	Uses Expectation-Maximization (EM), Maximum Likelihood Estimation (MLE)